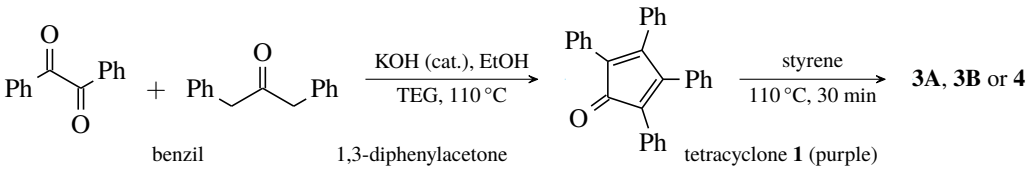
ORGANIC EXPERIMENT 2 — PRE-LAB CHECKLIST

One-pot Aldol Condensation and Diels–Alder Reaction

Condensed from the CM3191 Organic Lab Manual AY26/27 (Dr Hoang T. G., pp. 8–13) and lecture CM3191 Org L2. Week 8, **Thursday 8 October 2026**, Thursday lab. The official datasheet (manual p. 13) is attached as the last page of this document: print it, fill it in, get it signed.



1. Reagent table (manual values; blanks in the manual filled by calculation)

Reagent	MW / conc.	Density	Amount	mmol	Equiv.
Benzil (limiting)	210.23	—	0.21 g	1.00	1.00
1,3-Diphenylacetone	210.27	—	0.21 g	1.00	1.00
Triethylene glycol (TEG)	150.17	1.113	0.50 mL	solvent	—
KOH, 2 M in EtOH (fresh)	2 M	—	5–6 drops	cat.	—
Styrene (stabilised)	104.15	0.906	0.40 mL	3.48	3.5
Product 3A or 3B	488.63	—	**0.488 g (theor.)**	1.00	—
Product 4 (if CO lost)	460.62	—	**0.460 g (theor.)**	1.00	—

Theoretical yield uses your *actual* benzil mass: $m_{\text{theor}} = (m_{\text{benzil}}/210.23) \times \text{MW}_{\text{product}}$. Which MW applies is decided only after the structure is known (X-ray given later). Extra styrene if needed (1 equiv) ≈ 0.11 mL.

2. Bring / prepare before the session

☐ **Last page of this document printed** (official datasheet)
☐ **Ruler + pencil** (TLC distances in cm on the datasheet)
☐ Lab coat, goggles, nitrile gloves, covered shoes
☐ This bench sheet
☐ Phone for photos of every balance reading
☐ Labelled plastic bag for the product

3. Session timeline (estimate, not in the manual)

#	Stage	Approx.
1	Block to 110 °C; weigh solids; melt in TEG (bright yellow)	15 min
2	KOH drops → purple, solid; heat 3–5 min; TLC R1	20 min
3	Styrene, 110 °C **30 min** (+20 min if TLC R2 shows tetracyclone)	30–50 min
4	Cool, EtOH 5 mL, ice, Hirsch filtration, EtOH wash	20 min
5	Recrystallise (EtOH/EtOAc), RT 15–20 min, then ice	40 min
6	Filter, dry; mass, mp, IR, NMR sample, GTA signature	30 min

SAFETY — from the manual (p. 10)

All manipulations in the hood. **Styrene**: flammable, eye/skin irritant, inhalation hazard, IARC Group 2B (possibly carcinogenic) — gloves, goggles, hood only. Ethyl acetate, hexane, ethanol: flammable. KOH and benzil: skin/eye irritants, acutely toxic if swallowed. $CDCl_3$ and CH_2Cl_2: toxic by inhalation, suspected carcinogens. The product may release styrene if heated above 170 °C (relevant to the mp run up to 200 °C).

Added notes (not in the manual). The safety section also mentions iodine and oil baths, but the procedure uses a heating block and no iodine **[CONFIRM: ask TA]**. Heating block at 110 °C: handle the vial with tongs or a clamp. **Lesson from Exp. 1**: add the hot dissolving solvent (EtOAc) **dropwise**, never all at once.

BENCH MANUAL (ENGLISH)

One-pot Aldol Condensation and Diels–Alder Reaction

Tick each box as you go. "Why" notes explain the purpose; they are not extra steps.

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1. Aldol condensation

☐ 1. Switch on the heater with heating block, set **110 °C**. *Do this first.*

☐ 2. Into a **20 mL reaction vial**: **0.21 g benzil**, **0.21 g 1,3-diphenylacetone**, a small stir bar, **0.50 mL TEG**. Photograph both balance readings.

☐ 3. Heat on the block until the solids dissolve or melt: **bright yellow** solution.
Why: TEG has a very high bp and low volatility, so both steps can run in one hot vial without losing solvent.

☐ 4. Add **5–6 drops of fresh 2 M KOH/EtOH**. It should quickly turn **purple and solidify**; if not, add **2 more drops**. Heat **3–5 min**. Record the number of drops.
Why: base forms the enolate of 1,3-diphenylacetone; two aldol additions + two dehydrations give tetracyclone **1** (deep purple, extended conjugation). Only catalytic base is needed.

☐ 5. **TLC R1**: dip a pipette (no bulb) into the mixture, put the tip in a small vial, rinse it down with a little EtOAc from the bulb end. Spot **benzil | 1,3-diphenylacetone | R1**; elute with **10% EtOAc / 90% hexane**; view under UV; circle in pencil.
Identify the aldol product (the purple spot is visible even without UV). Starting materials gone → continue; otherwise consult the TA.

2. Diels–Alder reaction

☐ 6. Add **0.40 mL styrene** (hood, gloves). Heat **30 min at 110 °C**, stirring well. **Dark purple → tan**. Record the time.
Why: tetracyclone is the diene (locked s-*cis*), styrene the dienophile. As **1** is consumed, its colour fades. Efficient mixing speeds the reaction (manual).

☐ 7. **TLC R2**: sample as before, label R2. Spot **R1 | R2**; same eluent; UV.
If tetracyclone remains: add **1 equiv styrene (≈0.11 mL)** and heat **20 min** more. Record it.

3. Work-up

☐ 8. Cool to RT. Add **5 mL ethanol**, stir with a spatula. Cool in an **ice bath** with stirring: product precipitates.

☐ 9. **Suction filtration, Hirsch funnel**. Break up chunks with a metal spatula, spread out, **wash well with ethanol**.
Why: the product is poorly soluble in cold EtOH; the EtOH wash removes excess styrene.

4. Recrystallisation (mixed solvent)

☐ 10. Crude + stir bar in a **50 mL conical flask**; add **10 mL ethanol** (non-dissolving) + **10 mL ethyl acetate** (dissolving).

☐ 11. Second conical flask: ∼**20 mL ethyl acetate** + 2–3 boiling chips.

☐ 12. Heat both to **gentle boiling**. If solid remains, add hot EtOAc **dropwise** until it dissolves — **minimum**. Record the volume added.

☐ 13. Clear solution → cool undisturbed to RT **15–20 min**, then ice bath.

☐ 14. Suction filter, rinse with **minimal ethanol**, dry on the filter with vacuum on.
No crystals today: flask inside a beaker, cover with a watch glass, leave in your hood until the next session.

5. Analysis

☐ 15. Weigh in a **pre-weighed plastic bag**; photograph; calculate yield.

☐ 16. **Melting point**: Mel-Temp, start at **150 °C**, heat to **200 °C**, watch the whole time; record anything unusual (gas, colour change).

☐ 17. **IR** spectrum (note the C=O region). **NMR**: **10–20 mg** in $CDCl_3$ (gentle warming may be needed).

☐ 18. Show photos to the GTA; get the **printed datasheet signed**. X-ray structure is given later.

실험 매뉴얼 (한국어)

One-pot 알돌 축합 + Diels–Alder 반응

단계를 마칠 때마다 체크하세요. “이유”는 목적 설명이며 추가 단계가 아닙니다.

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1. 알돌 축합

☐ 1. 히팅 블록을 켜고 **110 °C**로 설정합니다. 가장 먼저 합니다.

☐ 2. **20 mL 바이알**에 **benzil 0.21 g**, **1,3-diphenylacetone 0.21 g**, 작은 교반 막대, **TEG 0.50 mL**를 넣습니다. 저울 값은 사진으로 남깁니다.

☐ 3. 블록에 올려 고체가 녹을 때까지 가열합니다. **밝은 노란색** 용액이 됩니다.
이유*:* TEG는 끓는점이 매우 높고 잘 날아가지 않아, 두 반응을 한 바이알에서 용매 손실 없이 진행할 수 있습니다.

☐ 4. **새로 만든 2 M KOH/EtOH를 5–6방울** 넣습니다. 곧 **보라색으로 변하며 굳어야** 합니다. 안 되면 **2방울 더**. **3–5분** 더 가열하고, 넣은 방울 수를 기록합니다.
이유*:* 염기가 1,3-diphenylacetone의 엔올레이트를 만들고, 알돌 첨가 두 번과 탈수 두 번으로 tetracyclone **1**(짙은 보라)이 생깁니다. 염기는 촉매량이면 충분합니다.

☐ 5. **TLC R1**: 고무 벌브 없는 피펫 끝을 혼합물에 담갔다 빼서 작은 바이알에 넣고, 반대쪽에서 EtOAc를 조금 흘려 녹입니다. **benzil | 1,3-diphenylacetone | R1** 순으로 찍고 **10% EtOAc/90% 헥산**으로 전개, UV에서 연필로 표시합니다.
보라색 점(알돌 생성물)은 UV 없이도 보입니다. 출발물질이 사라졌으면 다음 단계로, 아니면 TA와 상의합니다.

2. Diels–Alder 반응

☐ 6. **styrene 0.40 mL**를 넣고(후드, 장갑) **110 °C에서 30분**, 잘 교반합니다. **짙은 보라 → 황갈색**. 시각을 기록합니다.
이유*:* tetracyclone이 다이엔(s-*cis* 고정), styrene이 친다이엔체입니다. **1**이 소모되면서 보라색이 옅어집니다. 교반이 잘될수록 빨라집니다(매뉴얼).

☐ 7. **TLC R2**: 같은 방법으로 R2를 만들어 **R1 | R2**로 찍고 전개합니다.
tetracyclone이 남아 있으면 **styrene 1당량(약 0.11 mL)**을 더 넣고 **20분** 더 가열합니다. 기록하세요.

3. 후처리

☐ 8. 실온으로 식힌 뒤 **에탄올 5 mL**를 넣고 스패출러로 저어 줍니다. **얼음 수조**에서 저으면 생성물이 석출됩니다.

☐ 9. **Hirsch 깔때기로 감압 여과**. 덩어리는 금속 스패출러로 부수어 펴고, **에탄올로 충분히 씻습니다**.
이유*:* 생성물은 찬 에탄올에 잘 녹지 않고, 에탄올 세척이 남은 styrene을 제거합니다.

4. 재결정 (혼합 용매)

☐ 10. 조생성물과 교반 막대를 **50 mL 삼각플라스크**에 넣고 **에탄올 10 mL**(잘 안 녹이는 용매) + **에틸아세테이트 10 mL**(녹이는 용매)를 넣습니다.

☐ 11. 다른 삼각플라스크에 **에틸아세테이트 약 20 mL** + 끓임쪽 2–3개.

☐ 12. 둘 다 **약하게 끓을 때까지** 가열합니다. 고체가 남아 있으면 뜨거운 EtOAc를 **한 방울씩**, 녹을 만큼만 넣습니다. **넣은 양을 기록**합니다.

☐ 13. 맑아지면 흔들지 말고 실온에서 **15–20분**, 이어서 얼음 수조.

☐ 14. 감압 여과, **에탄올 최소량**으로 헹구고 진공을 건 채 거름종이 위에서 말립니다.
오늘 결정이 안 생기면: 플라스크를 비커 안에 넣고 시계접시로 덮어 다음 시간까지 후드에 둡니다.

5. 분석

☐ 15. **미리 무게를 잰 비닐봉투**에 담아 무게를 재고 사진, 수율 계산.

☐ 16. **녹는점**: Mel-Temp를 **150 °C**에서 시작해 **200 °C**까지, 끝까지 지켜보며 특이한 현상(기체, 색 변화)을 기록합니다.

☐ 17. **IR**(C=O 영역 확인). **NMR**: 생성물 **10–20 mg**을 $CDCl_3$에 녹입니다(살짝 데워야 할 수 있음).

☐ 18. 사진을 GTA에게 보여 주고 **인쇄한 datasheet에 서명**. X선 구조는 나중에 제공됩니다.

DATASHEET — WORKING LOG

One-pot Aldol Condensation and Diels–Alder Reaction

Name: KIM DONGHWAN Fumehood: _____ Date: 8 Oct 2026 Photos taken: ☐

Working log for your own use. The **official** datasheet is the last page: print it, get it signed, submit it with the report.

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1. Masses, amounts and analysis

Benzil (g)		1,3-Diphenylacetone (g)	
n(benzil) = mass/210.23 (mmol)		n(DPA) = mass/210.27 (mmol)	
KOH drops (total)		Styrene (mL) / extra styrene (mL)	
Empty plastic bag (g)		Bag + product (g)	
Product (g)		EtOAc added in recryst. (mL, est.)	
Theor. if **3A/3B**: $n \times 488.63$ (g)		Theor. if **4**: $n \times 460.62$ (g)	
Percentage yield (%) (fill in once the structure is known)			
Melting range (°C)		IR file name	
NMR tube no. ($CDCl_3$)		X-ray structure given	3A / 3B / 4

2. Time log

KOH added	Styrene added	Out of block	Crude filtered	Into ice (recryst.)

3. Checkpoints

Stage	**Expected** [source]	**Observed / time**	✓
Solids in TEG, 110 °C	dissolve or melt; bright yellow [M]		
After KOH	quickly purple and solid [M]; +2 drops if not [M]		
TLC R1	starting materials consumed [M]; purple spot of **1** [E]		
Styrene, 30 min	dark purple → tan [M]		
TLC R2	no tetracyclone left [M]; else +1 eq styrene, 20 min [M]		
EtOH 5 mL + ice	product precipitates [M]		
Hot EtOH/EtOAc	clear solution with minimum EtOAc [M]		
Cooling	crystals form; may need until next session [M]		
Dried product	white solid [M]		
mp 150→200 °C	record unusual observations [M]; styrene may be released above 170 °C [M]		

Sources: [M] stated in the lab manual; [E] general expectation, not in the manual. The Expected column is a **checkpoint, not an answer**: record what you actually see.

4. TLC (distances in cm; copy the drawing onto the official datasheet)

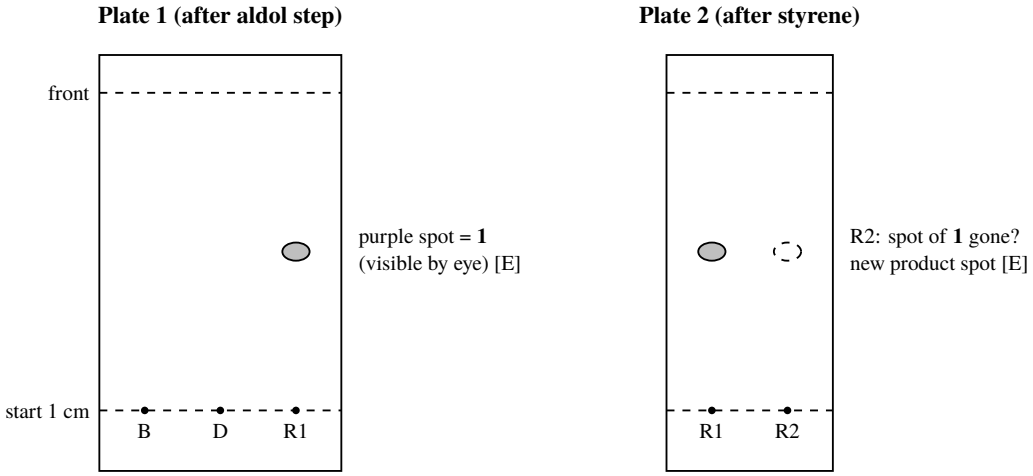
Plate / lane	**Spot (cm)**	**Front (cm)**	R_f	**Note (colour, UV)**
1: Benzil				
1: 1,3-Diphenylacetone				
1: R1				
2: R1				
2: R2				

5. Other observations

APPENDIX — TLC PLAN AND APPARATUS

What to spot, what to expect, how to record

Expectations marked [E] are general expectations, not statements from the manual; record what you actually see.

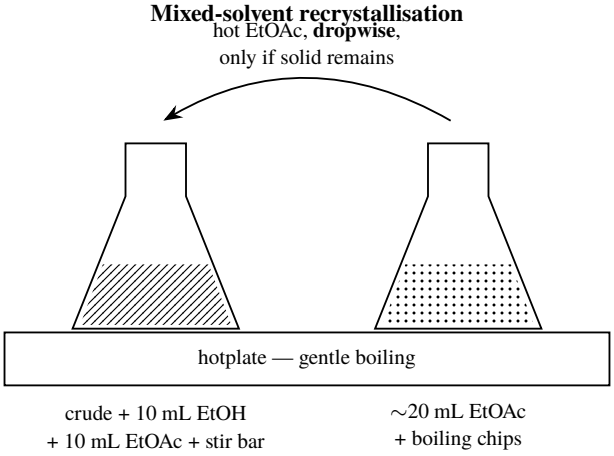


B = benzil, D = 1,3-diphenylacetone. Spot positions are schematic only. Eluent: 10% EtOAc / 90% hexane; UV lamp; circle every spot in pencil. **The datasheet asks you to draw the plate and label all distances in cm**: measure from the start line to the centre of each spot and to the solvent front.

$$R_f = \frac{\text{distance moved by the spot (start line to spot centre)}}{\text{distance moved by the solvent front}}$$

TLC tips (from the chromatography lecture)

- Pencil only; start line ∼1 cm from the bottom; solvent level in the jar below the start line.
- Small spots: touch the capillary briefly; let the solvent evaporate between applications.
- Saturate the jar (filter paper, lid on) before developing; take the plate out before the front reaches the top and mark the front immediately.
- Do not judge by colour alone: check every plate under UV.



WHAT TO RECORD FOR THE REPORT

Masses (photos) • number of KOH drops • colours at each stage (yellow → purple → tan) • TLC plates with distances (cm) • any extra styrene • volume of EtOAc added in the recrystallisation • mp range and anything seen while melting • IR and NMR file numbers.

Name: _____

Date of the experiment: _____

Datasheet for One pot Aldol Condensation and Diels-Alder Reaction

Instruction: Take photos of the measurements. After completing the experiment, show the photos to your teaching assistant and ask them to sign your printed datasheet (hardcopy, not softcopy). Submit the signed datasheet along with your lab report

Starting materials

- Actual mass of benzil used (g)_____
- Actual mass of 1,3-diphenylacetone used (g)_____
- Amount of styrene used (mL)_____

Product

- Mass of empty plastic bag (g)_____
- Mass of plastic bag + product (g) _____
- Mass of your final product (g) _____
- Percentage yield (%): _____

Observation during the experiment:

Draw your TLC plate here and label all the distances (in cm)